

Operating Systems
15/03/2020
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6. How many processes will be created by the code fragment below, excluding the initial parent process?

```
if (fork() || fork()) {
    fork();
}
```

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7. How many processes are created by the code fragment below, when the parent process P calls f(3)?

```
void f(int n) {
    if (n > 0 || fork() == 0) {
        f(n-1);
        exit(0);
    }
    wait(0);
}
```

 ∞
 does not stop

8. What will the code fragment below print to the console?

```
char* s[3] = {"A", "B", "C"};
for (i=0; i<3; i++) {
    if (fork() == 0) {
        execl("/bin/echo", "/bin/echo", s[i], NULL);
    }
}
```

 ABCCBCC

9. What does the system call "read" do when the pipe contains less data than it is required to read, but it is not empty?
 It reads existing data

10. What will the code fragment below print to the console?

```
int p[2];
char buf[10];
int n;
pipe(p);
n = read(p[0], buf, 10);
printf("%d\n", n);
```

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11. Why are zombie processes problematic?
 They occupy memory. Their parent is taking time.
 They are not doing anything but they are taking time.

12. Consider that function f is executed simultaneously by 10 threads. Add the necessary lines of code to insure that the value of n is 10 after the threads have completed?

```
int n = 0;
pthread_mutex_t m;
void* f(void* p) {
    pthread_lock(&m);
    n++;
    pthread_unlock(&m);
    return NULL;
}
```

13. Schedule the following jobs (given as Name/Duration/Deadline) so that the sum of their delays is minimized: A/7/13, B/5/9, C/2/4
 CBA
 delay = 14/13

File name: _____
 Username: _____
 Email: _____
 Date: _____

1. Write a shell script command that displays the name of the user who is logged in, the name of the machine he is logged in (e.g., 10.0.0.1), and the name of the terminal (e.g., 10.0.0.1).

grep 'logname' /etc/passwd

2. Write a shell script command that displays the name of the user who is logged in, the name of the machine he is logged in (e.g., 10.0.0.1), and the name of the terminal (e.g., 10.0.0.1).

cat /etc/passwd | cut -d: -f1,2,3

3. Write a shell script command that displays the name of the user who is logged in, the name of the machine he is logged in (e.g., 10.0.0.1), and the name of the terminal (e.g., 10.0.0.1).

cat /etc/passwd | cut -d: -f1,2,3

4. Given a directory and all its sub-directories, write a shell script command that displays all sub-directories containing a file.

find . -type d -exec ls -la {} \;

5. Write a shell script that calculates the average number of lines of code in all the files with extension ".c" in the current directory, ignoring comment lines, empty lines, and the lines containing only spaces and tabs.

lines=0 total=0
for dir in \$(find . -type c)

do
total=\$((total + \$(wc -l < \$dir)))

lines=\$((lines + 1))
done

echo \$total

Operating Systems Examination 2018-2019, 2 = Test 2

Name: _____
Surname: _____
Grade: _____

1. Write a UNIX Shell command that displays all the lines in file a.txt that contain at least a distance in kilometers (i.e. 5 km).

`grep "[0-9]\+ km\+" a.txt`

2. Write a UNIX Shell command that displays the lines of file a.txt swapping any pairs of upper case vowel followed by even digit (i.e. a23E7i2B4U -> a23E7i2B4U)

`sed "s/([AEIOU])([0-9]\+)/\2\1/g" a.txt`

3. Write a UNIX Shell command that displays a list of all unique appearances of SCS usernames belonging to the English or Romanian sections (e.g. abiel234, cdire5678) that appear in a text file a.txt.

`grep -o "[^[:punct:]]+[a-z][a-z]ic[0-9]\+" | uniq`

4. Given directory and all its hierarchy, write a UNIX Shell command that displays all sub-directories containing .c files.

`find . -type d -exec find {} -type f -name "*.c" \;`

5. Write a UNIX Shell script that calculates the average number of lines of code in all the files with extension ".sh" in the current directory, ignoring comment lines, empty lines, and the lines containing only spaces and tabs.

`lines=0 total=0`
`for fm in $(find . -type f -name "*.sh"); do`

`total=$((total+$(wc -l < $fm)))`

`lines=$((lines+1))`

`done`
`avg=$((total/lines))`
`echo $avg`

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```

6

7. How many processes are created by the code fragment below, when the parent process P calls f(3)?

```
void f(int n) {
    if(n > 0 || fork() == 0) {
        f(n-1);
        exit(0);
    }
    wait(0);
}
```

∞
does not stop

8. What will the code fragment below print to the console?

```
char* s[3] = {"A", "B", "C"};
for(i=0; i<3; i++) {
    if(fork() == 0) {
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ABCCBCC

9. What does the system call "read" do when the pipe contains less data than it is required to read, but it is not empty?

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char buf[10];
int m;
pipe(p);
n = read(p[0], buf, 10);
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```

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11. Why are zombie processes problematic?

They occupy memory. Their parent is PID=1 and clearing them wastes time.

12. Consider that function f is executed simultaneously by 10 threads. Add the necessary lines of code to insure that the value of n is 10 after the threads have completed?

```
int n = 0;
pthread_mutex_t m;
```

```
void* f(void* p) {
    pthread_lock(&m);
    n++;
    pthread_unlock(&m);
    return NULL;
}
```

13. Schedule the following jobs (given as Name/Duration/Deadline) so that the sum of their delays is minimised: A/7/13, B/3/9, C/2/4

CBA

delay=1 14/13

14. Give an advantage and a disadvantage of the set-associative caches versus the direct caches.

Set-associative avoids collision, is flexible.
(advantage)

Set-associative does not use all available cache lines (disadvantage)

15. What page has the highest priority in the LRU replacement policy, when choosing a victim page?

$N \times N$ matrix
Pages with the minimum number of 1's on its line.

16. Given two set-associative caches, one with 2 sets of 4 pages and one with 4 sets of 2 pages, which would perform better for the following sequence of page requests: 20, 9, 18, 27, 20, 9, 18, 27. Why?

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They perform the same. Replacements/transfers are not needed. The one with 2-sets has to iterate by one to find 18 and 27 place respectively. 4 sets performs better.
$$\text{page-address} \% \text{cache-sets} = \text{set of page}$$

17. How many data blocks can be referenced to by the triple-indirection of an i-node, if a block contains N addresses to other blocks?

N^3

18. Consider the producer-consumer problem with a buffer of capacity N. How many semaphores would you use to insure operation correctness and what would be the semaphores' initial values?

N semaphores with initial values 0.

19. Give a method for preventing the operation of deadlock, in a situation when you cannot avoid modifying resources concurrently.

Pick an order for resources and always look in that order.

20. What is a binary semaphore, and what is the effect of its P method, when called by multiple concurrent processes/threads?

A binary semaphore has only two values 1 and 0. If value is 0, process is put in queue, else it is put in ~~run~~runned